Fall 2004, CIS, Temple University

CIS527: Data Warehousing, Filtering, and Mining

Lecture 2

- Data Warehousing and OLAP Technology for Data Mining

Lecture slides taken/modified from:

- Jiawei Han (http://www-sal.cs.uiuc.edu/~hanj/DM_Book.html)

Chapter 2: Data Warehousing and OLAP Technology for Data Mining

- What is a data warehouse?
- A multi-dimensional data model
- Data warehouse architecture
- Data warehouse implementation
- Further development of data cube technology
- From data warehousing to data mining

What is Data Warehouse?

- Defined in many different ways, but not rigorously.
  - A decision support database that is maintained separately from the organization's operational database
  - Support information processing by providing a solid platform of consolidated, historical data for analysis.
- "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon
- Data warehousing:
  - The process of constructing and using data warehouses

Data Warehouse—Subject-Oriented

- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process.
- Organized around major subjects, such as customer, product, sales.
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing.

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources
  - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
    - E.g., Hotel price: currency, tax, breakfast covered, etc.
  - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems.
  - Operational database: current value data.
  - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
  - Contains an element of time, explicitly or implicitly
  - However, the key of operational data may or may not contain "time element".

Data Warehouse—Non-Volatile

- A physically separate store of data transformed from the operational environment.
- Operational update of data does not necessarily occur in the data warehouse environment.
  - Does not require transaction processing, recovery, and concurrency control mechanisms
  - Often requires only two operations in data accessing:
    - *initial loading of data* and *access of data*.

Data Warehouse vs. Heterogeneous DBMS

- Traditional heterogeneous DB integration:
  - Build wrappers/mediators on top of heterogeneous databases
  - Query driven approach
    - A query posed to a client site is translated into queries appropriate for individual heterogeneous sites; The results are integrated into a global answer set
    - Involving complex information filtering
    - Competition for resources at local sources
- Data warehouse: update-driven, high performance
  - Information from heterogeneous sources is integrated in advance and stored in warehouses for direct query and analysis

Data Warehouse vs. Operational DBMS

- OLTP (on-line transaction processing)
  - Major task of traditional relational DBMS
  - Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- OLAP (on-line analytical processing)
  - Major task of data warehouse system
  - Data analysis and decision making
- Distinct features (OLTP vs. OLAP):
  - User and system orientation: customer vs. market
  - Data contents: current, detailed vs. historical, consolidated
  - Database design: ER + application vs. star + subject
  - View: current, local vs. evolutionary, integrated
  - Access patterns: update vs. read-only but complex queries

Why Separate Data Warehouse?

- High performance for both systems
  - DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
  - Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation.
- Different functions and different data:
  - Decision support requires historical data which operational DBs do not typically maintain
  - Decision Support requires consolidation (aggregation, summarization) of data from heterogeneous sources
  - Different sources typically use inconsistent data representations, codes and formats which have to be reconciled

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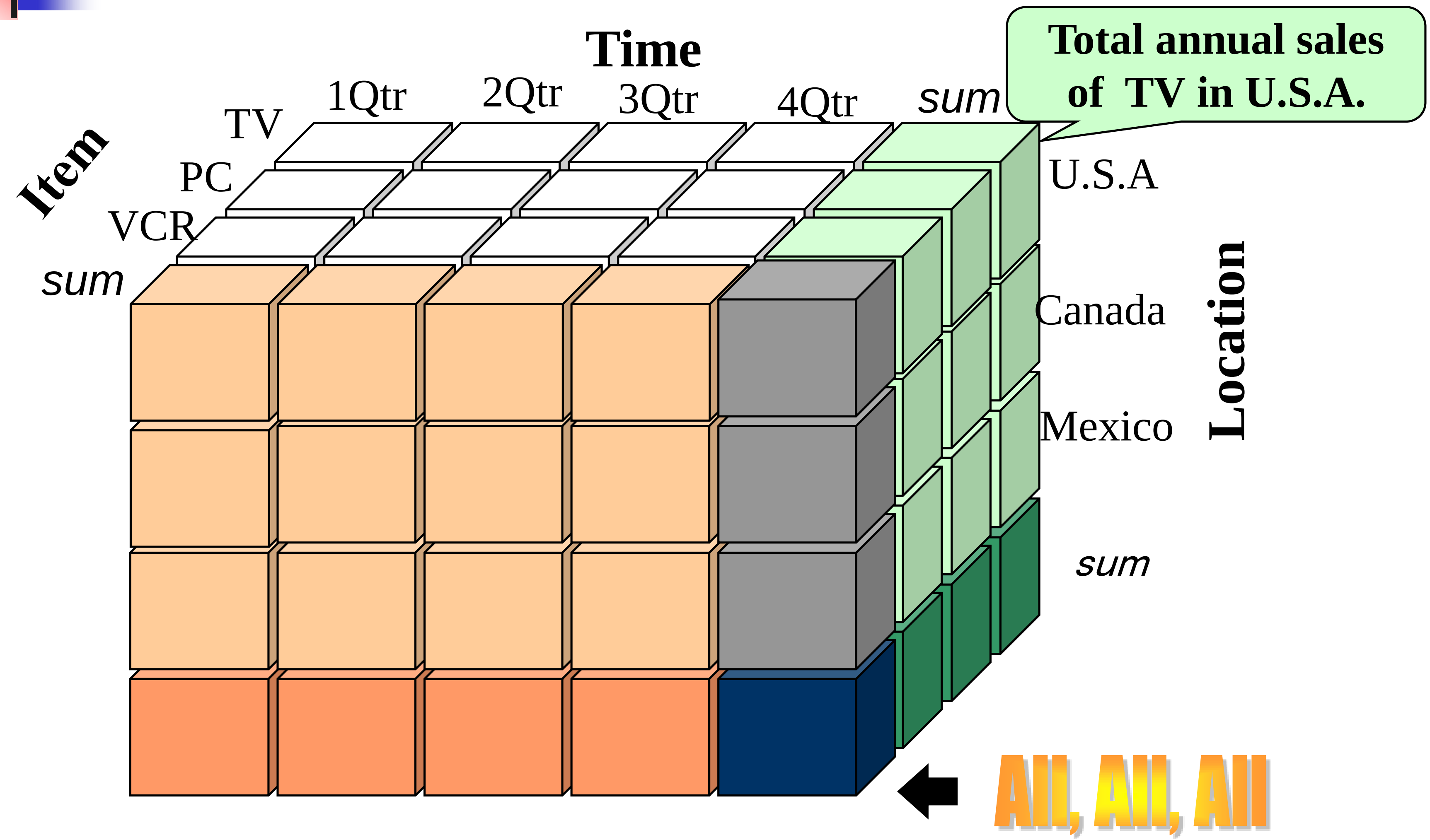
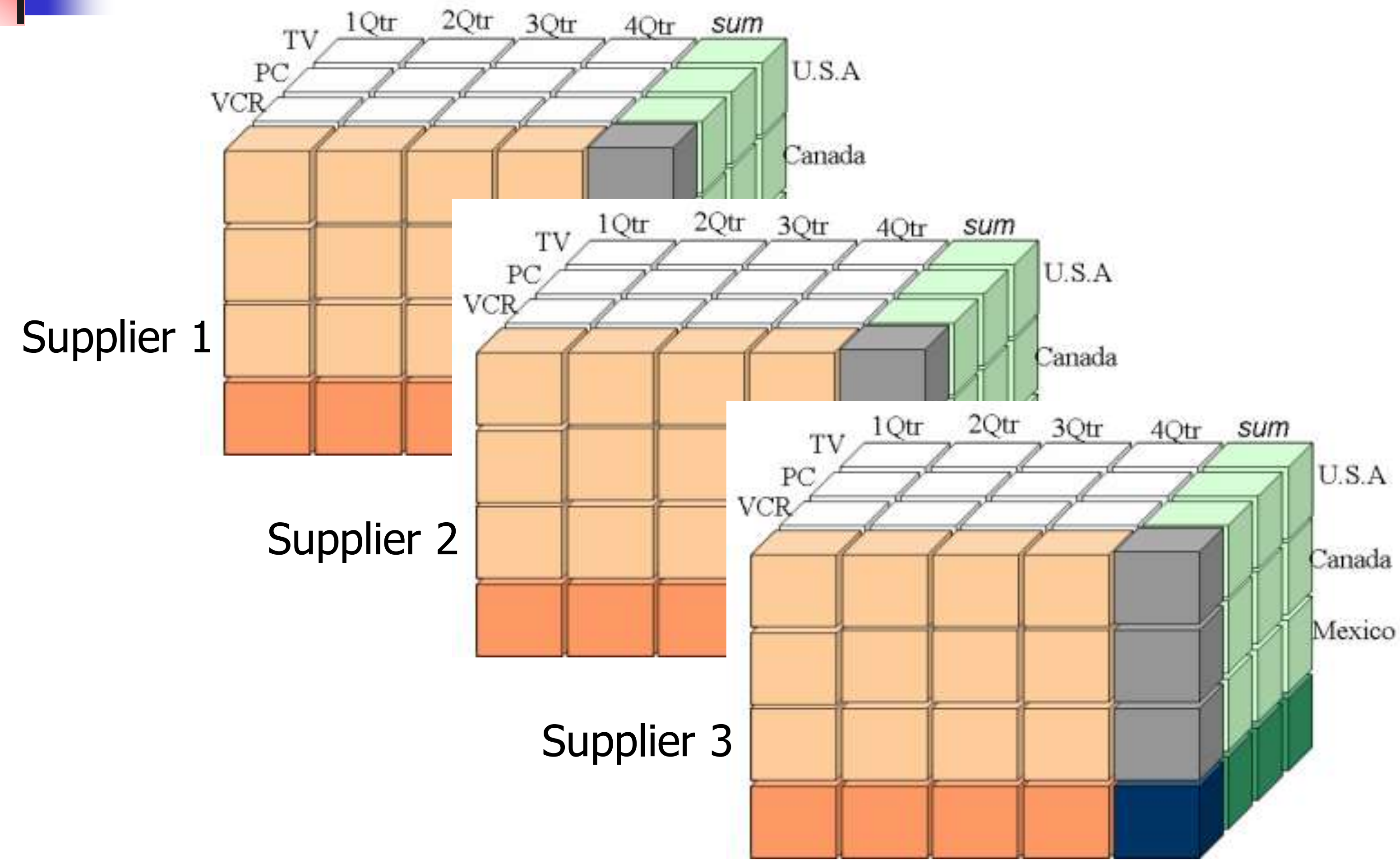
A Multi-Dimensional Data Model

- A data warehouse is based on a multidimensional data model which views data in the form of a data cube
- A data cube allows data to be modeled and viewed in multiple dimensions
  - Dimension tables, such as item (item_name, brand, type), or time(day, week, month, quarter, year)
  - Fact table contains measures (such as dollars_sold) and keys to each of the related dimension tables
- In data warehousing literature, an n-D base cube is called a base cuboid. The top most 0-D cuboid, which holds the highest-level of summarization, is called the apex cuboid. The lattice of cuboids forms a data cube.

A Sample Data Cube

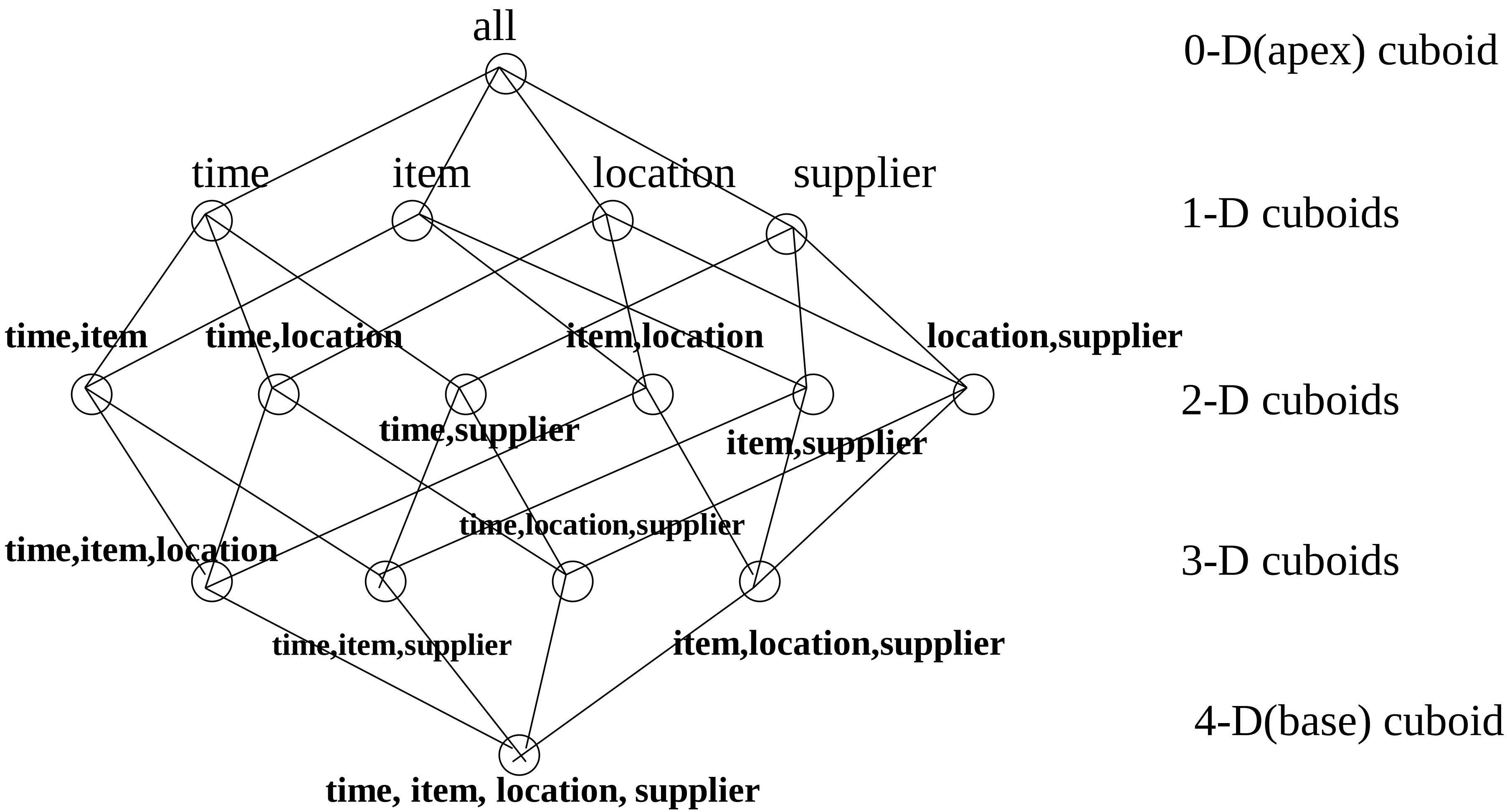


4-D Data Cube



TV
PC
VCR
1Qtr
2Qtr
3Qtr
4Qtr
sum
U.S.A
Canada
Supplier 1
TV
PC
VCR
1Qtr
2Qtr
3Qtr
4Qtr
sum
U.S.A
Canada
Supplier 2
TV
PC
VCR
1Qtr
2Qtr
3Qtr
4Qtr
sum
U.S.A
Canada
Mexico
Supplier 3

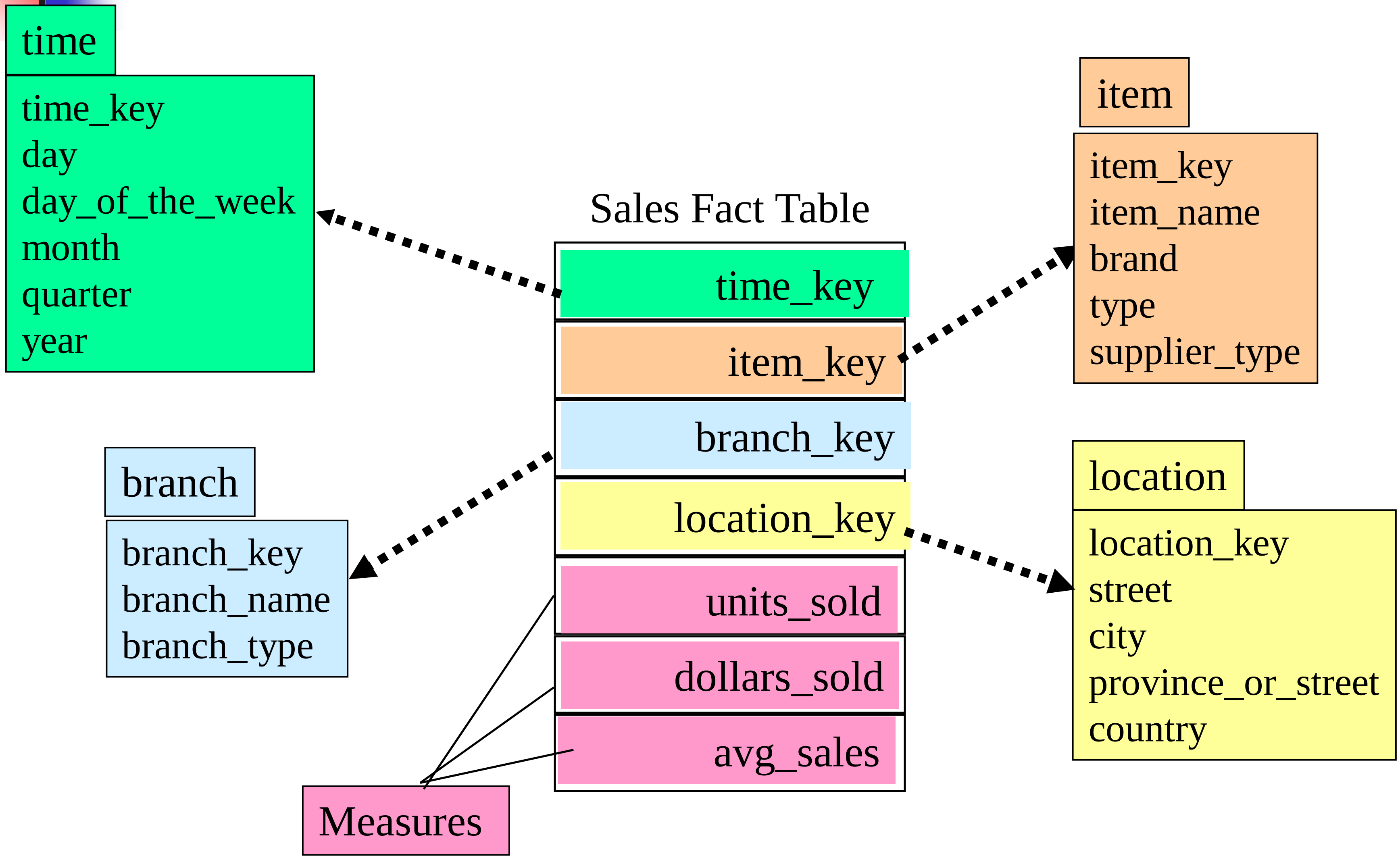
Cube: A Lattice of Cuboids



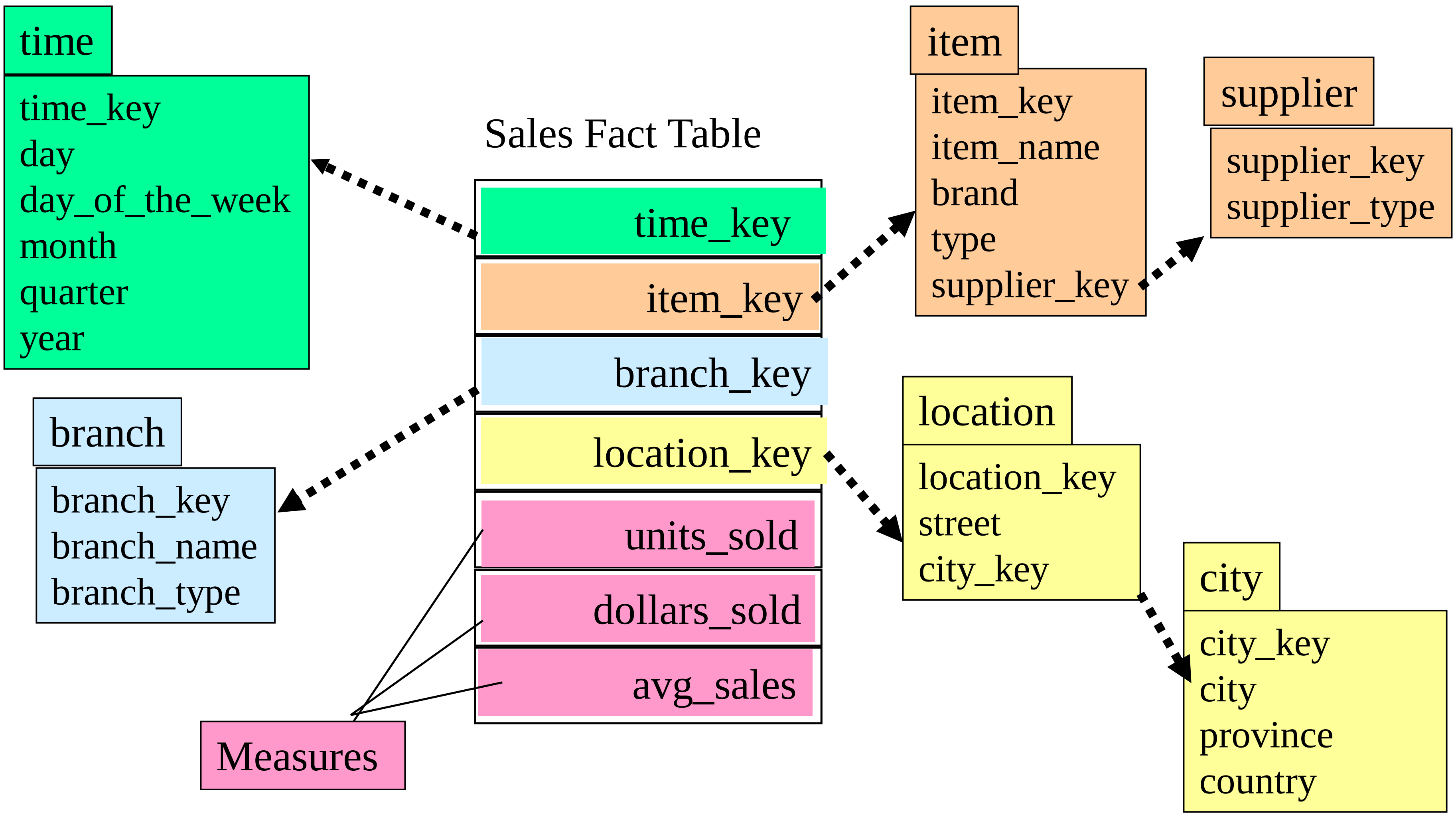
Conceptual Modeling of Data Warehouses

- Modeling data warehouses: dimensions & measures
  - Star schema: A fact table in the middle connected to a set of dimension tables
  - Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
  - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

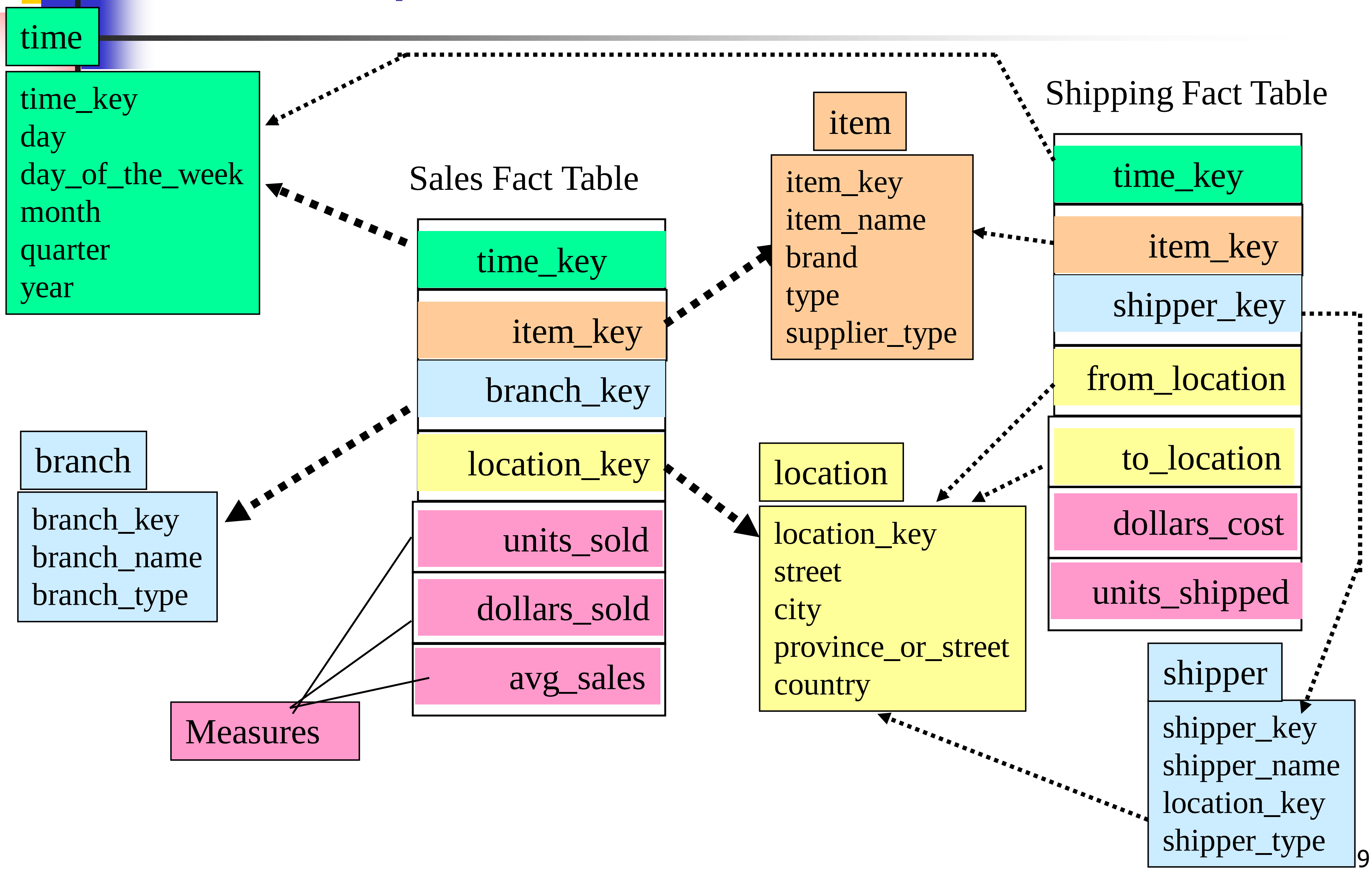
Example of Star Schema



Example of Snowflake Schema



Example of Fact Constellation



A Data Mining Query Language, DMQL: Language Primitives

- Cube Definition (Fact Table)

  define cube <cube_name> [<dimension_list>]: <measure_list>
- Dimension Definition (Dimension Table)

  define dimension <dimension_name> as (<attribute_or_subdimension_list>)
- Special Case (Shared Dimension Tables)
  - First time as "cube definition"
  - define dimension <dimension_name> as <dimension_name_first_time> in cube <cube_name_first_time>

Defining a Star Schema in DMQL

define cube sales_star [time, item, branch, location]:

dollars_sold = sum(sales_in_dollars), avg_sales = avg(sales_in_dollars), units_sold = count(*)

define dimension time as (time_key, day, day_of_week, month, quarter, year)

define dimension item as (item_key, item_name, brand, type, supplier_type)

define dimension branch as (branch_key, branch_name, branch_type)

define dimension location as (location_key, street, city, province_or_state, country)

Defining a Snowflake Schema in DMQL

define cube sales_snowflake [time, item, branch, location]:

dollars_sold = sum(sales_in_dollars), avg_sales = avg(sales_in_dollars), units_sold = count(*)

define dimension time as (time_key, day, day_of_week, month, quarter, year)

define dimension item as (item_key, item_name, brand, type, supplier(supplier_key, supplier_type))

define dimension branch as (branch_key, branch_name, branch_type)

define dimension location as (location_key, street, city(city_key, province_or_state, country))

Defining a Fact Constellation in DMQL

```
define cube sales [time, item, branch, location]:
        dollars_sold = sum(sales_in_dollars), avg_sales =
          avg(sales_in_dollars), units_sold = count(*)
define dimension time as (time_key, day, day_of_week, month, quarter, year)
define dimension item as (item_key, item_name, brand, type, supplier_type)
define dimension branch as (branch_key, branch_name, branch_type)
define dimension location as (location_key, street, city, province_or_state,
   country)
define cube shipping [time, item, shipper, from_location, to_location]:
        dollar_cost = sum(cost_in_dollars), unit_shipped = count(*)
define dimension time as time in cube sales
define dimension item as item in cube sales
define dimension shipper as (shipper_key, shipper_name, location as location
   in cube sales, shipper_type)
define dimension from_location as location in cube sales
define dimension to_location as location in cube sales
```

Measures: Three Categories

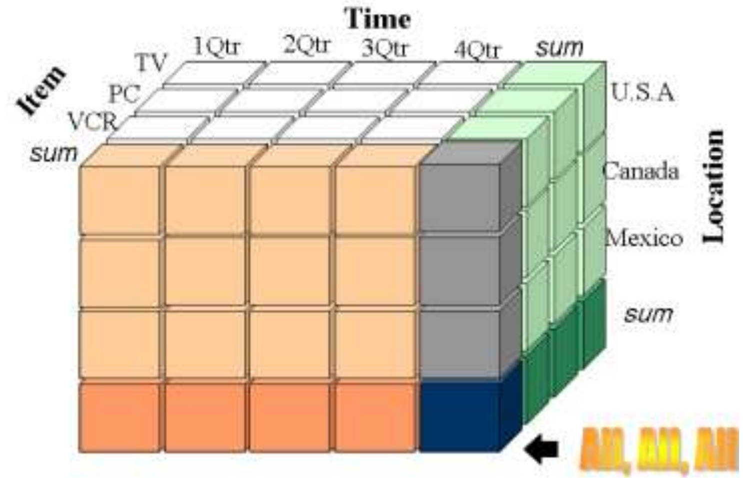
Measure: a function evaluated on aggregated data corresponding to given dimension-value pairs.

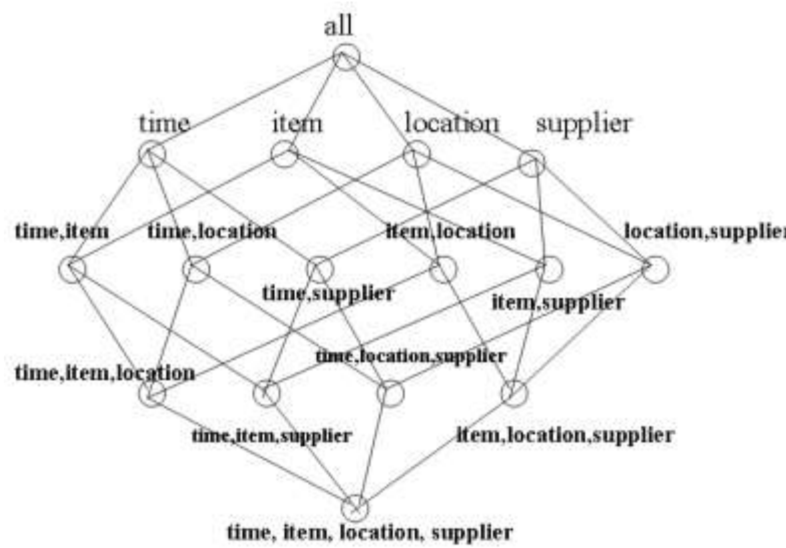
Measures can be:

- distributive: if the measure can be calculated in a distributive manner.
  - E.g., count(), sum(), min(), max().
- algebraic: if it can be computed from arguments obtained by applying distributive aggregate functions.
  - E.g., avg()=sum()/count(), min_N(), standard_deviation().
- holistic: if it is not algebraic.
  - E.g., median(), mode(), rank().

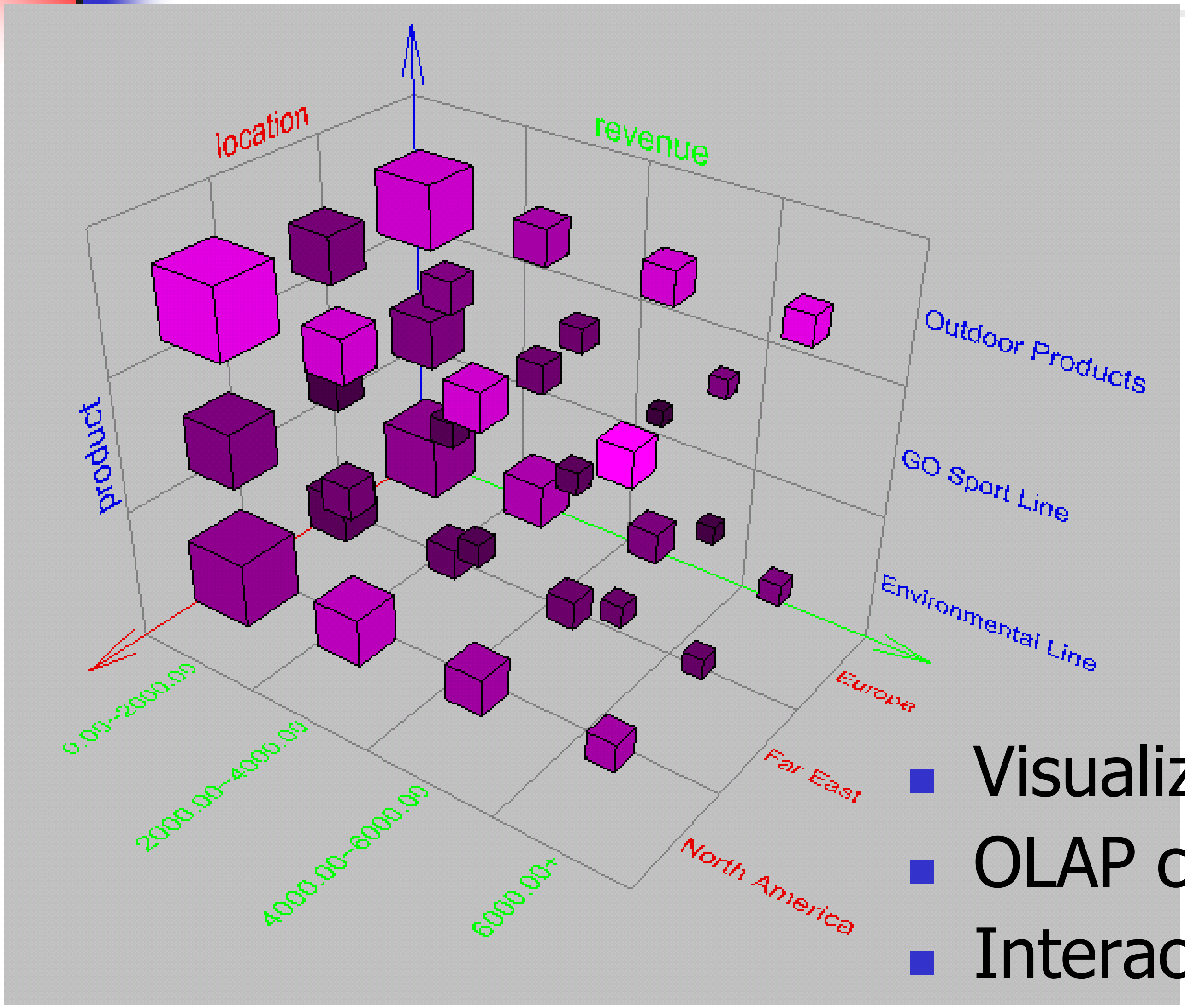
Measures: Three Categories

- Distributive and algebraic measures are ideal for data cubes.
- Calculated measures at lower levels can be used directly at higher levels.
- Holistic measures can be difficult to calculate efficiently.
- Holistic measures could often be efficiently approximated.





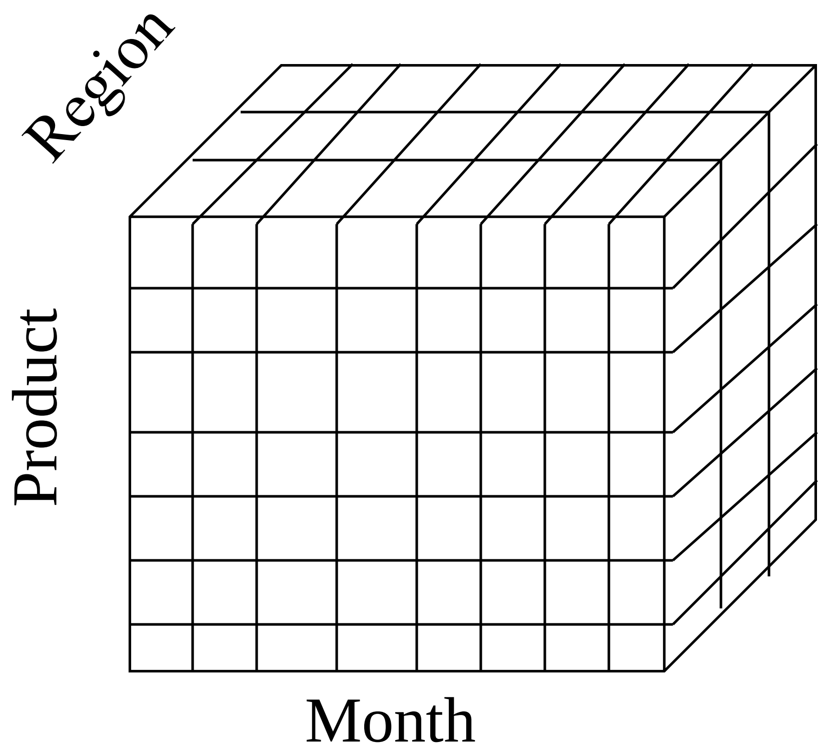
Browsing a Data Cube



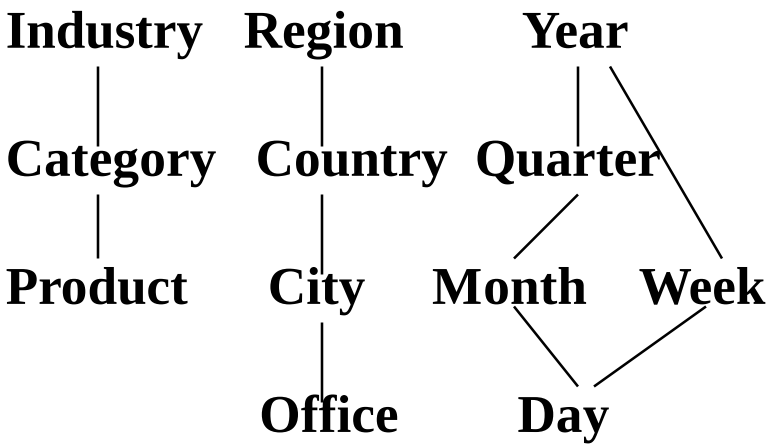
- Visualization
- OLAP capabilities
- Interactive manipulation

A Concept Hierarchy

- Concept hierarchies allow data to be handled at varying levels of abstraction



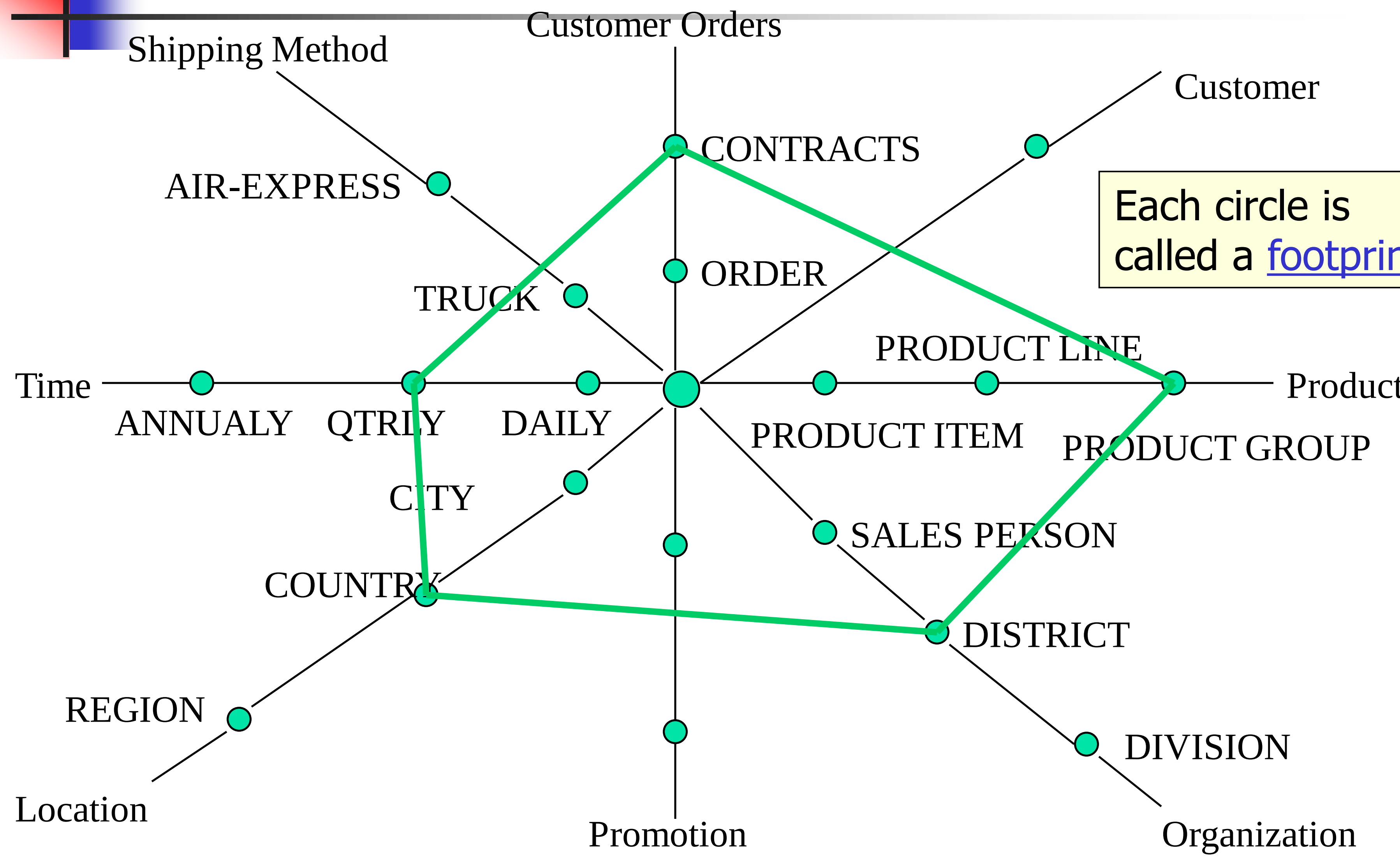
Dimensions: Product, Location, Time
Hierarchical summarization paths



Typical OLAP Operations (Fig 2.10)

- Roll up (drill-up): summarize data
  - *by climbing up concept hierarchy or by dimension reduction*
- Drill down (roll down): reverse of roll-up
  - *from higher level summary to lower level summary or detailed data, or introducing new dimensions*
- Slice and dice:
  - *project and select*
- Pivot (rotate):
  - *reorient the cube, visualization, 3D to series of 2D planes.*
- Other operations
  - *drill across: involving (across) more than one fact table*
  - *drill through: through the bottom level of the cube to its back-end relational tables (using SQL)*

Querying Using a Star-Net Model



Each circle is called a footprint

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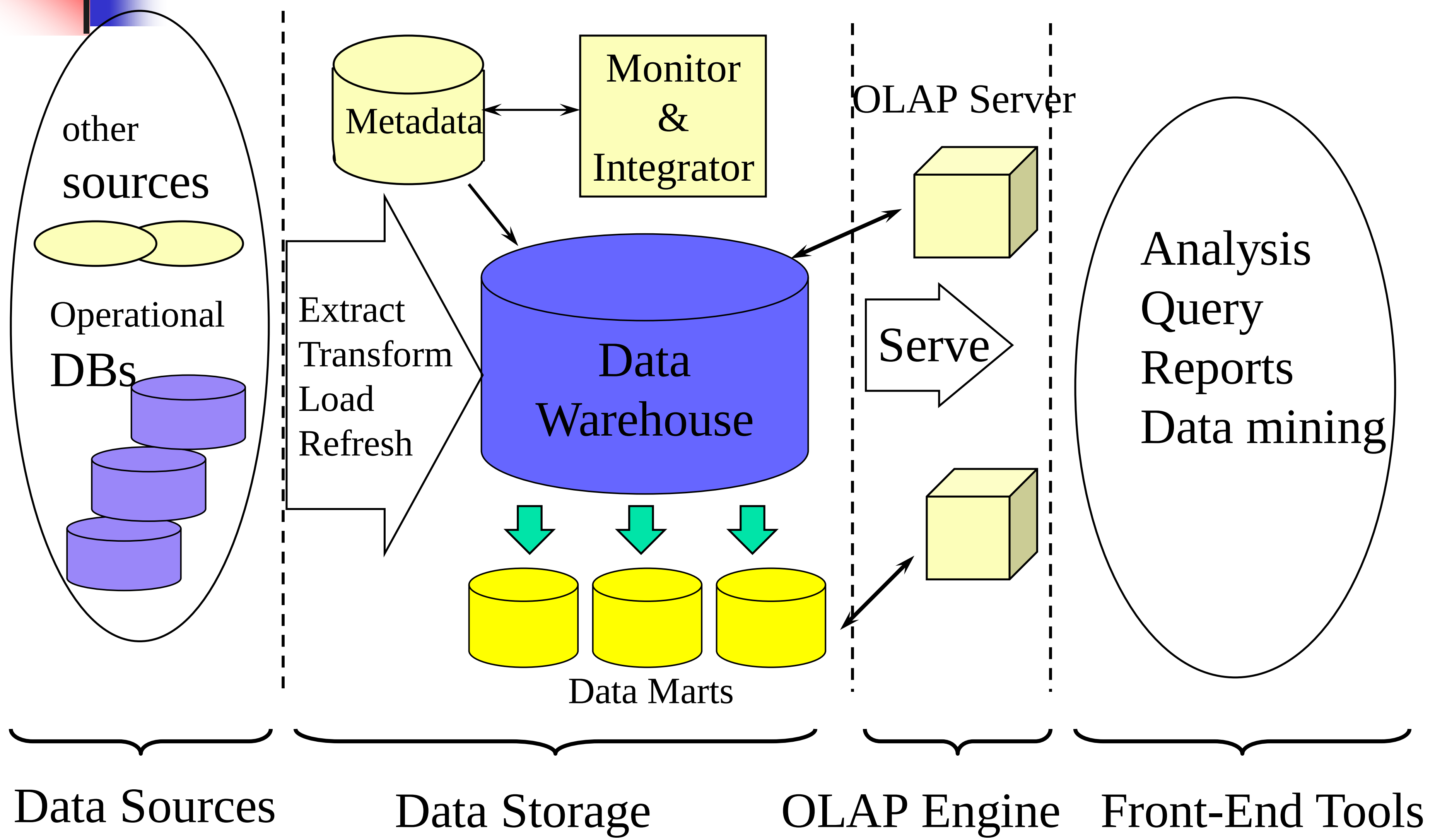
Data Warehouse Design Process

- Top-down, bottom-up approaches or a combination of both
  - Top-down: Starts with overall design and planning (mature)
  - Bottom-up: Starts with experiments and prototypes (rapid)
- From software engineering point of view
  - Waterfall: structured and systematic analysis at each step before proceeding to the next
  - Spiral: rapid generation of increasingly functional systems, quick modifications, timely adaptation of new designs and technologies
- Typical data warehouse design process
  - Choose a business process to model, e.g., orders, invoices, etc.
  - Choose the *grain* (*atomic level of data*) of the business process
  - Choose the dimensions that will apply to each fact table record
  - Choose the measure that will populate each fact table record

Multi-Tiered Architecture



Three Data Warehouse Models

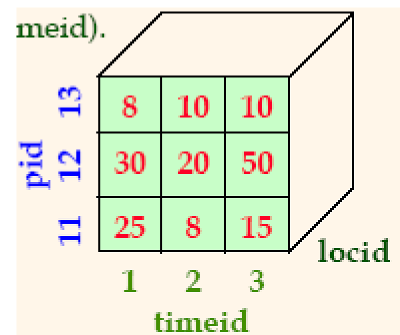
- Enterprise warehouse
  - collects all of the information about subjects spanning the entire organization
- Data Mart
  - a subset of corporate-wide data that is of value to a specific groups of users. Its scope is confined to specific, selected groups, such as marketing data mart
    - Independent vs. dependent (directly from warehouse) data mart
- Virtual warehouse
  - A set of views over operational databases
  - Only some of the possible summary views may be materialized

OLAP Server Architectures

- Relational OLAP (ROLAP)
  - Use relational or extended-relational DBMS to store and manage warehouse data
  - Include optimization of DBMS backend and additional tools and services
  - greater scalability
- Multidimensional OLAP (MOLAP)
  - Array-based multidimensional storage engine (sparse matrix techniques)
  - fast indexing to pre-computed summarized data
- Hybrid OLAP (HOLAP)
  - User flexibility (low level: relational, high-level: array)
- Specialized SQL servers
  - specialized support for SQL queries over star/snowflake schemas

pid	timei	locid	sales
11	1	1	25
11	2	1	8
11	3	1	15
12	1	1	30
12	2	1	20
12	3	1	50
13	1	1	8
13	2	1	10
13	3	1	10
11	1	2	35

- - -



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Efficient Data Cube Computation

- Data cube can be viewed as a lattice of cuboids
  - The bottom-most cuboid is the base cuboid
  - The top-most cuboid (apex) contains only one cell
  - How many cuboids in an n-dimensional cube with L levels?

$$T = \prod_{i=1}^{n} (L_i + 1)$$

- Materialization of data cube
  - Materialize every (cuboid) (full materialization), none (no materialization), or some (partial materialization)
  - Selection of which cuboids to materialize
    - Based on size, sharing, access frequency, etc.

Cube Operation

- Cube definition and computation in DMQL

  define cube sales[item, city, year]: sum(sales_in_dollars)

  compute cube sales

- Transform it into a SQL-like language (with a new operator cube by, introduced by Gray et al.'96)

  SELECT item, city, year, SUM (amount)

  FROM SALES

  CUBE BY item, city, year

- Need compute the following Group-Bys

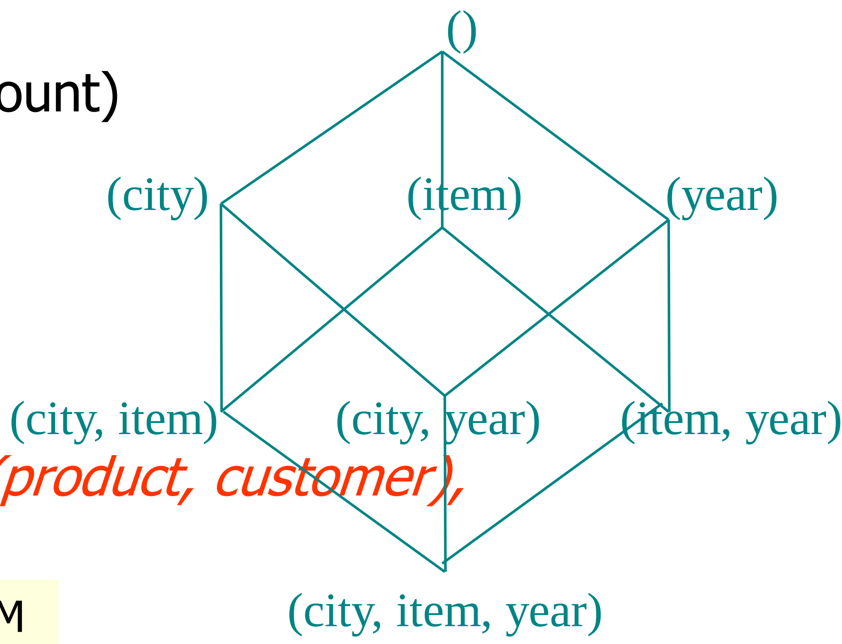
  (date, product, customer),

  (date,product),(date, customer), (product, customer),

  (date), (product), (customer)

  ()

SELECT item, city, year, SUM (amount)
FROM SALES
GROUP BY item, year

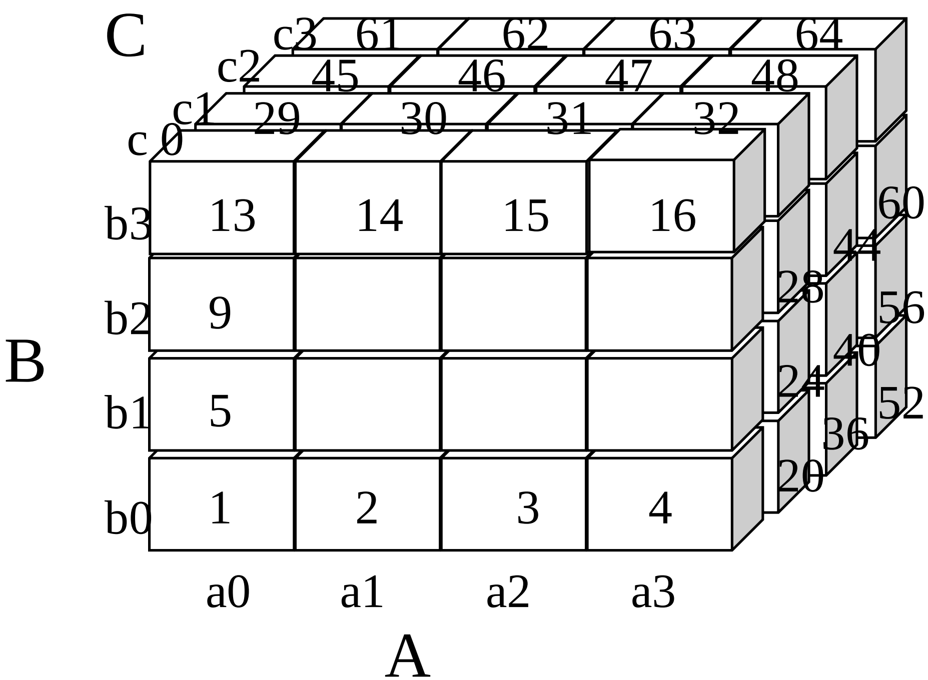


Cube Computation: ROLAP vs. MOLAP

- ROLAP-based cubing algorithms
  - Key-based addressing
  - Sorting, hashing, and grouping operations are applied to the dimension attributes to reorder and cluster related tuples
  - Aggregates may be computed from previously computed aggregates, rather than from the base fact table
- MOLAP-based cubing algorithms
  - Direct array addressing
  - Partition the array into chunks that fit the memory
  - Compute aggregates by visiting cube chunks
  - Possible to exploit ordering of chunks for faster calculation

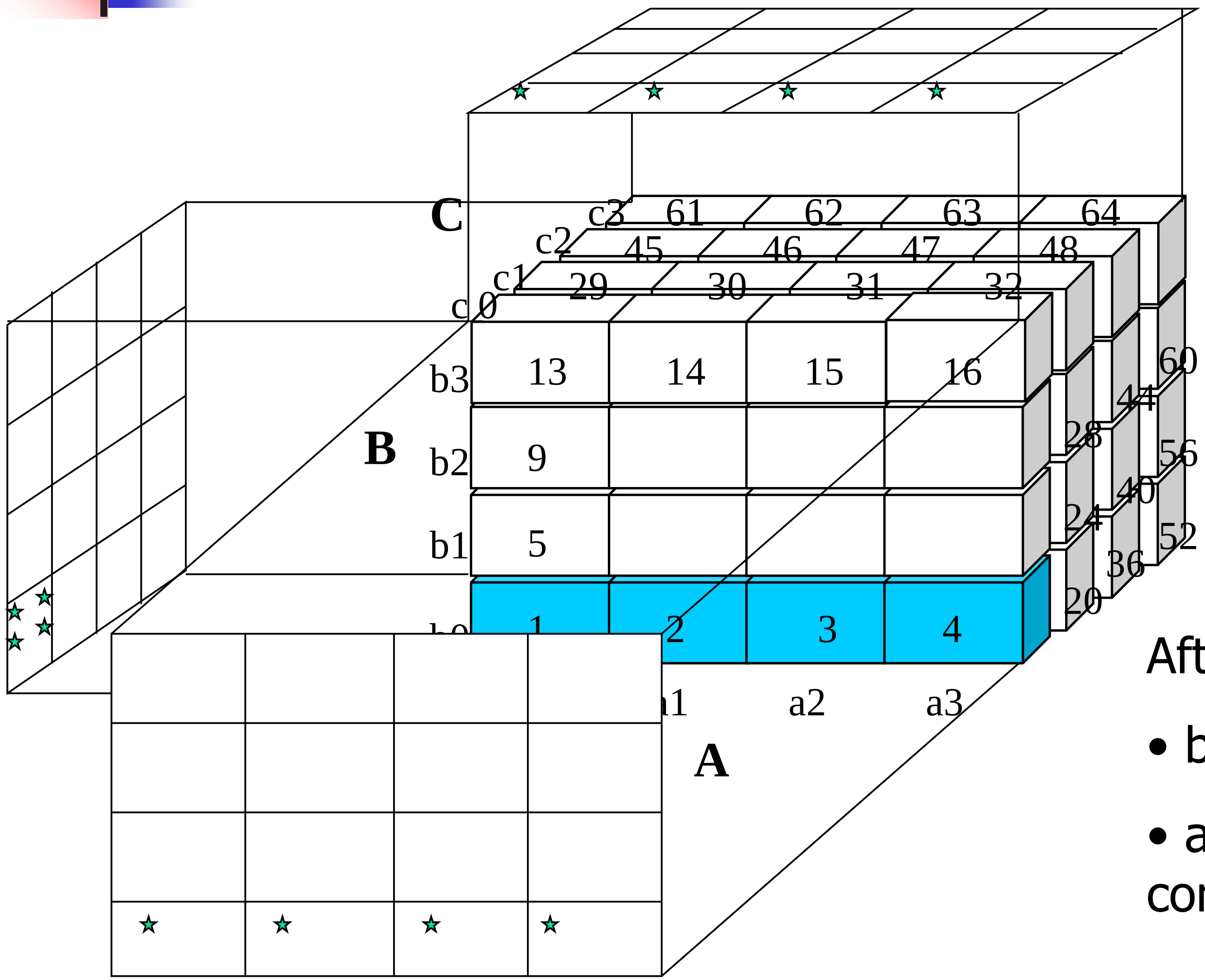
Multiway Array Aggregation for MOLAP

- Partition arrays into chunks (a small subcube which fits in memory).
- Compressed sparse array addressing: (chunk_id, offset)
- Compute aggregates in "multiway" by visiting cube cells in the order which minimizes the # of times to visit each cell, and reduces memory access and storage cost.



What is the best traversing order to do multi-way aggregation?

Multiway Array Aggregation for MOLAP



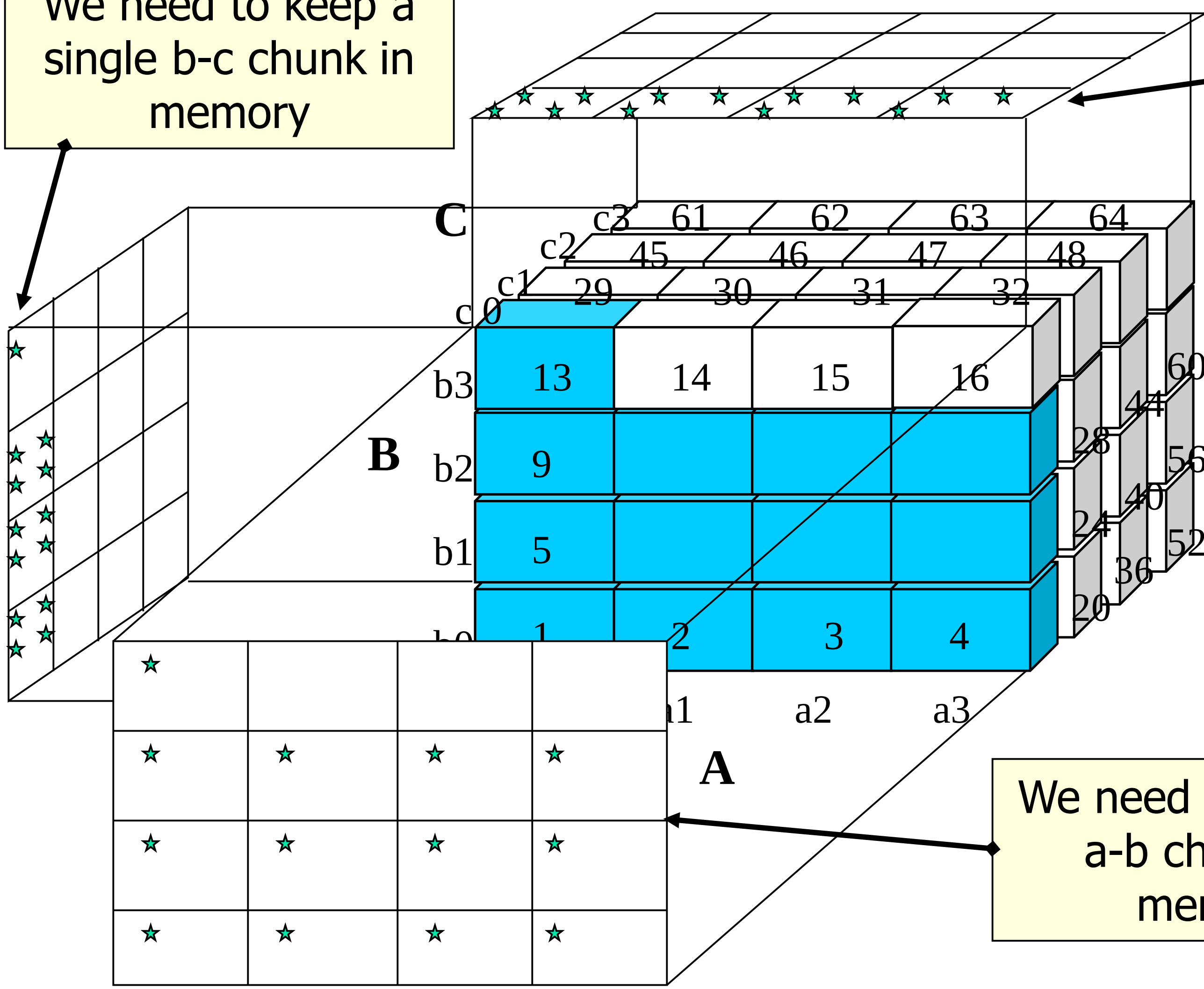
After scan {1,2,3,4}:

- b_0c_0 chunk is computed
- a_0c_0 and a_0b_0 are not computed

Multiway Array Aggregation for MOLAP

We need to keep a single b-c chunk in memory

We need to keep 4 a-c chunks in memory



After scan 1-13:

- a_0c_0 and b_0c_0 chunks are computed
- a_0b_0 is not computed (we will need to scan 1-49)

We need to keep 16 a-b chunks in memory

Multiway Array Aggregation for MOLAP

- Method: the planes should be sorted and computed according to their size in ascending order.
  - The proposed scan is optimal if |C|>|B|>|A|
  - See the details of Example 2.12 (pp. 75-78)
- MOLAP cube computation is faster than ROLAP
- Limitation of MOLAP: computing well only for a small number of dimensions
- If there are a large number of dimensions use the iceberg cube computation: process only "dense" chunks

Indexing OLAP Data: Bitmap Index

- Suitable for low cardinality domains
- Index on a particular column
- Each value in the column has a bit vector: bit-op is fast
- The length of the bit vector: # of records in the base table
- The *i*-th bit is set if the *i*-th row of the base table has the value for the indexed column

Base table

Cust	Region	Type
C1	Asia	Retail
C2	Europe	Dealer
C3	Asia	Dealer
C4	America	Retail
C5	Europe	Dealer

Index on Region

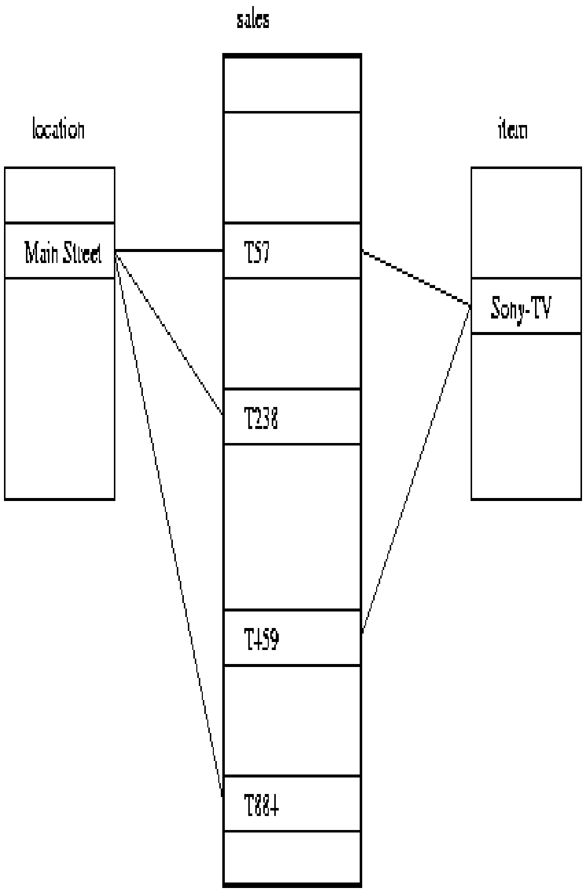
RecID	Asia	Europe	America
1	1	0	0
2	0	1	0
3	1	0	0
4	0	0	1
5	0	1	0

Index on Type

RecID	Retail	Dealer
1	1	0
2	0	1
3	0	1
4	1	0
5	0	1

Indexing OLAP Data: Join Indices

- Join index materializes relational join and speeds up relational join — a rather costly operation
- In data warehouses, join index relates the values of the dimensions of a start schema to rows in the fact table.
  - E.g. fact table: *Sales* and two dimensions *location* and *item*
    - A join index on *location* is a list of pairs <loc_name,T_id> sorted by location
    - A join index on *location-and-item* is a list of triples <loc_name,item_name, T_id> sorted by location and item names
- Search of a join index can still be slow
- Bitmapped join index allows speed-up by using bit vectors instead of dimension attribute names



Online Aggregation

- Consider an aggregate query:

  "finding the average sales by state"

- Can we provide the user with some information before the exact average is computed for all states?
  - Solution: show the current "running average" for each state as the computation proceeds.
  - Even better, if we use statistical techniques and sample tuples to aggregate instead of simply scanning the aggregated table, we can provide bounds such as "the average for Wisconsin is 2000±102 with 95% probability.

Efficient Processing of OLAP Queries

- Determine which operations should be performed on the available cuboids:
  - transform drill, roll, etc. into corresponding SQL and/or OLAP operations, e.g, dice = selection + projection
- Determine to which materialized cuboid(s) the relevant operations should be applied.
- Exploring indexing structures and compressed vs. dense array structures in MOLAP (trade-off between indexing and storage performance)

Metadata Repository

- Meta data is the data defining warehouse objects. It has the following kinds
  - Description of the structure of the warehouse
    - schema, view, dimensions, hierarchies, derived data definitions, data mart locations and contents
  - Operational meta-data
    - data lineage (history of migrated data and transformation path), currency of data (active, archived, or purged), monitoring information (warehouse usage statistics, error reports, audit trails)
  - The algorithms used for summarization
  - The mapping from operational environment to the data warehouse
  - Data related to system performance
    - warehouse schema, view and derived data definitions
  - Business data
    - business terms and definitions, ownership of data, charging policies

Data Warehouse Back-End Tools and Utilities

- Data extraction:
  - get data from multiple, heterogeneous, and external sources
- Data cleaning:
  - detect errors in the data and rectify them when possible
- Data transformation:
  - convert data from legacy or host format to warehouse format
- Load:
  - sort, summarize, consolidate, compute views, check integrity, and build indices and partitions
- Refresh
  - propagate the updates from the data sources to the warehouse

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Discovery-Driven Exploration of Data Cubes

- Hypothesis-driven: exploration by user, huge search space
- Discovery-driven (Sarawagi et al.'98)
  - pre-compute measures indicating exceptions, guide user in the data analysis, at all levels of aggregation
  - Exception: significantly different from the value anticipated, based on a statistical model
  - Visual cues such as background color are used to reflect the degree of exception of each cell
  - Computation of exception indicator can be overlapped with cube construction

Examples: Discovery-Driven Data Cubes

item	all
region	all

Sum of sales	month											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total		1%	-1%	0%	1%	3%	-1	-9%	-1%	2%	-4%	3%

Avg sales	month											
item	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sony b/w printer		9%	-8%	2%	-5%	14%	-4%	0%	41%	-13%	-15%	-11%
Sony color printer		0%	0%	3%	2%	4%	-10%	-13%	0%	4%	-6%	4%
HP b/w printer		-2%	1%	2%	3%	8%	0%	-12%	-9%	3%	-3%	6%
HP color printer		0%	0%	-2%	1%	0%	-1%	-7%	-2%	1%	-5%	1%
IBM home computer		1%	-2%	-1%	-1%	3%	3%	-10%	4%	1%	-4%	-1%
IBM laptop computer		0%	0%	-1%	3%	4%	2%	-10%	-2%	0%	-9%	3%
Toshiba home computer		-2%	-5%	1%	1%	-1%	1%	5%	-3%	-5%	-1%	-1%
Toshiba laptop computer		1%	0%	3%	0%	-2%	-2%	-5%	3%	2%	-1%	0%
Logitech mouse		3%	-2%	-1%	0%	4%	6%	-11%	2%	1%	-4%	0%
Ergo-way mouse		0%	0%	2%	3%	1%	-2%	-2%	-5%	0%	-5%	8%

item	IBM home computer

Avg sales	month											
region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
North		-1%	-3%	-1%	0%	3%	4%	-7%	1%	0%	-3%	-3%
South		-1%	1%	-9%	6%	-1%	-39%	9%	-34%	4%	1%	7%
East		-1%	-2%	2%	-3%	1%	18%	-2%	11%	-3%	-2%	-1%
West		4%	0%	-1%	-3%	5%	1%	-18%	8%	5%	-8%	1%

Chapter 2: Data Warehousing and OLAP Technology for Data Mining

- What is a data warehouse?
- A multi-dimensional data model
- Data warehouse architecture
- Data warehouse implementation
- Further development of data cube technology
- From data warehousing to data mining

Data Warehouse Usage

- Three kinds of data warehouse applications
  - Information processing
    - supports querying, basic statistical analysis, and reporting using crosstabs, tables, charts and graphs
  - Analytical processing
    - multidimensional analysis of data warehouse data
    - supports basic OLAP operations, slice-dice, drilling, pivoting
  - Data mining
    - knowledge discovery from hidden patterns
    - supports associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools.
- Differences among the three tasks

From On-Line Analytical Processing to On Line Analytical Mining (OLAM)

- Why online analytical mining?
  - High quality of data in data warehouses
    - DW contains integrated, consistent, cleaned data
  - Available information processing structure surrounding data warehouses
    - ODBC, OLEDB, Web accessing, service facilities, reporting and OLAP tools
  - OLAP-based exploratory data analysis
    - mining with drilling, dicing, pivoting, etc.
  - On-line selection of data mining functions
    - integration and swapping of multiple mining functions, algorithms, and tasks.
- Architecture of OLAM

Summary

- Data warehouse
  - A subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process
- A multi-dimensional model of a data warehouse
  - Star schema, snowflake schema, fact constellations
  - A data cube consists of dimensions & measures
- OLAP operations: drilling, rolling, slicing, dicing and pivoting
- OLAP servers: ROLAP, MOLAP, HOLAP
- Efficient computation of data cubes
  - Partial vs. full vs. no materialization
  - Multiway array aggregation
  - Bitmap index and join index implementations
- Further development of data cube technology
  - Discovery-drive and multi-feature cubes
  - From OLAP to OLAM (on-line analytical mining)